Sovin Herzlinger

Pomona & Saratoga, CA ● sovinh.27@gmail.com
Portfolio: sherzlinger.github.io

EDUCATION

California State Polytechnic University, Pomona **August 2023 – Expected Dec 2027**
Bachelor of Science in Electromechanical Systems Engineering Technology (EMSET)
Relevant coursework: Instrumentation and Control · Mechatronics · Introduction to Autonomous Vehicles · Machine Elements

WORK EXPERIENCE

Early-Stage Hardware Startup **April 2025 – Present**
Systems Engineer
- Co-built the full electromechanical system of a precision hardware product from the ground up, as one of two electromechanical engineers on a small team.
- Wrote Python auto-tuning software for the product's PID control loops, replacing manual gain tuning.
- Built and equipped the engineering laboratory from scratch, sourcing all equipment and coordinating bids for specialized manufacturing.
- Designed and executed test campaigns and presented technical results to stakeholders.

Transcutaneous Spinal Stimulation Research **February 2026 – Present**
Researcher | Casa Colina Hospital, Pomona, CA
- Measure isometric force output in stroke patients with a strength testing rig to establish baseline motor function.
- Co-developing a second-generation transcutaneous spinal stimulation device targeting specific wrist movements, with a USC PhD researcher and two other students.
- Next phase, with patient trials planned for Summer 2027, applies transcutaneous spinal cord stimulation (tSCS) along the motor cortex-to-arm pathway to test whether it rebuilds the neuromuscular connection and restores strength.

PROJECTS AND VENTURES

Single-Photon Optical Comms Drone — Senior Project (National Research Lab) **Fall 2026 – Present**
- Building a free-space laser link between drone gimbals, based on single-photon research at a national research lab: two 850 nm lasers combined 99:1, the strong beam clocking an FPGA and the weak beam carrying pulse-position-encoded data.
- Own the pointing controls (RTK GPS coarse, NIR beacon medium, quadrant-photodiode fine tracking) and lead planning across gimbal, aiming, and optical-bench workstreams.

Linear Inverted Pendulum — Self-Balancing Cart-Pole (Controls Lab) **March 2026 – May 2026**
- Designed and built a belt-reduced cart-pole on a 750 mm rail driven by an ODrive S1 with an AS5600 pivot encoder, on a CAN bus across two Teensy 4.x nodes and the ODrive separating sensing, control, and motor drive.
- Implemented a 600 Hz LQR controller that catches and holds the pendulum upright from a hand-raised start, with live gain tuning over USB serial.
- Wrote a Python tool that solves the Riccati equation on the linearized plant to generate LQR gains; hardware-tuned gains landed within 30% of the model, damping within 0.2%.

AEON — Self-Hosted Personal AI Assistant **2024 – Present**
- Built a self-hosted assistant on a Raspberry Pi 5 (16 GB) with a FastAPI backend, reachable from all my devices over a private network with no public exposure; it manages my calendar, tasks, notes, and health data and writes a daily briefing to an e-ink reader.
- Fine-tuned Qwen3-4B with QLoRA (rank 32, ~1.6% of weights trained) to handle calendar and task requests locally; it answers 92% of a 250-request blind test set correctly in 1.6 s on a laptop.
- Personal data stays on my own hardware: calendar and task tools run locally and are never exposed to the cloud model used for open-ended chat. Shipped a companion iOS app to TestFlight.

QuickCheck — Campus Club Check-In Platform **June 2025 – Present**
- Founded and built a check-in, membership, and dues platform for campus organizations, now a native iOS app in external TestFlight beta; onboarded and managed a contracted engineer, handling contract negotiation, legal review, and scope.
- Rebuilt the product natively in React Native / Expo, migrating from Firebase to a Supabase Postgres backend and enforcing authorization through row-level security policies in the database rather than in application code.
- Ran a security review on the live database and fixed every high-severity finding, verified by testing the policies as a real signed-in user.

SKILLS

Mechanical & Simulation: SolidWorks | Onshape | Ansys HFSS | 3D Printing | Fabrication | Soldering | Prototyping
Controls & Embedded: LQR | PID Auto-Tuning | State-Space | MATLAB | CAN Bus | I^2C | BLE | Teensy | ESP32 / ESP-IDF | Raspberry Pi | Jetson | ODrive | C++
Software & AI: Python | TypeScript | React Native | Swift | PostgreSQL | Supabase | FastAPI | Docker | Ollama | QLoRA Fine-Tuning | Linux | Git